

The empirical recurrence for OEIS A236519: recovering a conjecture that is not written down

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Abstract

OEIS A236519 records “Empirical recurrence of order 82”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 101$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236519 is “Number of $(n+1) \times (4+1)$ 0..1 arrays with the difference of the upper median and minimum value of each 2×2 subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.”. It has offset 1 and begins

476, 5670, 63208, 655460, 6591144, 63997604, 610630403, 5723134951, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:19:13, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 313$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 101$ of the 313, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 101$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 626$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	15	c_{22}	−1285952201848417	c_{43}	1792759845540503285	c_{64}	4724526462880978
c_2	97	c_{23}	−1708836775801200	c_{44}	17290039026365107806	c_{65}	−1124536422866365
c_3	−2275	c_{24}	7125148499642727	c_{45}	−2620424818891463931	c_{66}	−794549344013549
c_4	−2037	c_{25}	7261365855448669	c_{46}	−15734392375121248390	c_{67}	179509621081518
c_5	155675	c_{26}	−32856802745788637	c_{47}	2904024945784131865	c_{68}	108685172751196
c_6	−145042	c_{27}	−25400179822943039	c_{48}	12251778567377836701	c_{69}	−22961967258632
c_7	−6438208	c_{28}	126967375970387835	c_{49}	−2576891813902606176	c_{70}	−11897206392680
c_8	11799344	c_{29}	72945204992060269	c_{50}	−8155560288022759777	c_{71}	2309383112396
c_9	181487688	c_{30}	−413379630048110392	c_{51}	1873584386394719595	c_{72}	1019530585708
c_{10}	−430202365	c_{31}	−170529698174327095	c_{52}	4634300774607283087	c_{73}	−177778899905
c_{11}	−3723410352	c_{32}	1138864380017947208	c_{53}	−1128533574018040227	c_{74}	−66320236681
c_{12}	10179132426	c_{33}	318121003524446690	c_{54}	−2243325754259543568	c_{75}	10065673791
c_{13}	57914049786	c_{34}	−2664125871570287990	c_{55}	566031197167015056	c_{76}	3128682980
c_{14}	−174106885700	c_{35}	−451038047142803607	c_{56}	922511981511015084	c_{77}	−393179398
c_{15}	−701931505822	c_{36}	5306108077435449021	c_{57}	−236782460283627526	c_{78}	−99478419
c_{16}	2269029198322	c_{37}	415027344911069916	c_{58}	−321109990872015670	c_{79}	9418990
c_{17}	6758456198442	c_{38}	−9016672415837002202	c_{59}	82539359366644646	c_{80}	1863556
c_{18}	−23255738199273	c_{39}	−25752019302627042	c_{60}	94173602046325706	c_{81}	−103736
c_{19}	−52415004682870	c_{40}	13092964755276858683	c_{61}	−23904868602229429	c_{82}	−14880
c_{20}	191477974279710	c_{41}	−776370177872300208	c_{62}	−23133715576830956		
c_{21}	330692258264623	c_{42}	−16263188941097705709	c_{63}	5723300591318288		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82 + 4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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